Automatic Control Project

Directions:

- A simulation using **MATLAB** is a must for this project.
- Plot the responses got by MATLAB.
- Soft copies of the solution are required.
- Solution must be in groups of **3 students**, not more.
- The deadline is 09/5/2026.

Steps:

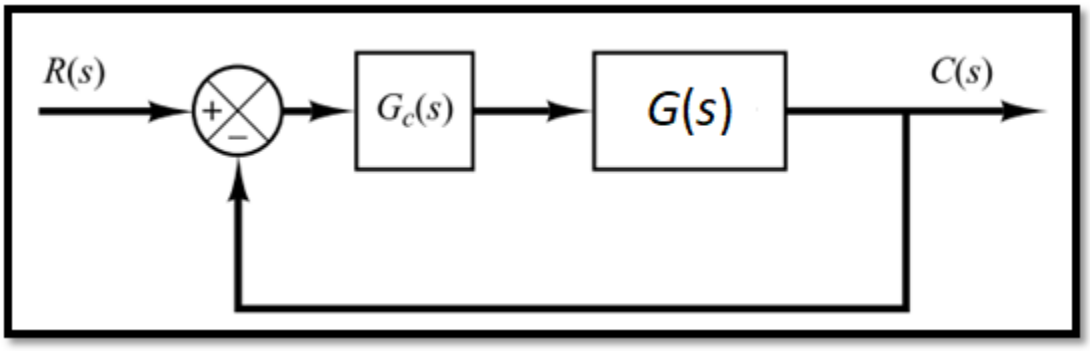
For the forward transfer function given:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Where ω_n is the last 2 numbers in the smallest ID in the group, and ζ is equal (0.6 if ω_n is multiple of 5, 0.25 if ω_n is multiple of 3, else ζ is equal 0.75.

You are required to do the following:

- Derive the closed loop transfer function (unity feedback) with the values of components given above assuming unity feedback.
- Use MATLAB to get the Step response, Impulse response, Rise time, Peak time,Settling time, Maximum peak response, and Root loci of this system.
- Now, it's required to design a compensator $G_c(s)$ to improve the system response as shown:



- Use MATLAB to design a suitable compensator to get a maximum peak of approximately ***20%*** and settling time less than or equal 2sec.
- Derive the system closed loop transfer function after inserting the compensator.
- Analyze the system after inserting the compensator in the feed forward path for both transient and steady state response using MATLAB.
- Design an analog circuit that represents the compensator.